

Evaluation Awareness Is Not One Capability: Evidence from Open Language Models

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Abstract

Safety benchmarks assume that test-condition behavior predicts deployment behavior, an assumption that fails if models detect evaluation cues and adapt. This opens a gap between benchmark performance and deployment behavior: compliance measured under test conditions becomes an optimistic upper bound that overstates how safely a model behaves once the evaluation harness is removed. We characterize this *evaluation awareness* through eight experiments across 37 open-weight models and seven families. (i) Detection is moderate and training-driven (24/37 models exceed chance, best AUROC 0.714 vs. 0.819 human, with instruction tuning dominating over scale). (ii) Detection shifts safety behavior (hard refusal drops 5.8 pp under hypothetical framing, and 21/140 Harm-Bench framing effects are significant, with compliance rising up to +30 pp). (iii) Representations survive behavioral collapse (probes retain AUROC ≥ 0.98 under rewrites that drive behavior below chance, and multi-layer steering causally moves three downstream tasks while random controls do not). (iv) These axes are weakly coupled (only 1/15 correlations are significant, the sole robust link being behavioral detection versus framing resistance, $\rho = -0.79$, $p < 0.001$). We call this gap the **benchmark illusion**: because detectability, behavioral manifestation, and controllability vary independently, it is multivariate rather than a single number, so no single awareness score is a reliable proxy for deployment safety.

1 Introduction

Safety benchmarks for large language models (LLMs) rest on a silent assumption: that behavior measured under test conditions is a reliable proxy for behavior in deployment. This assumption fails if models can detect test-like cues and shift their behavior in response. This creates a practical problem for model governance: benchmark scores become optimistic upper bounds, reassuring numbers that

overstate how safely a model will act once the test harness is removed.

Current frontier research acknowledges that *evaluation awareness*, the ability of a trained model to recognize that it is being evaluated, is a real phenomenon (Needham et al., 2025; Abdelnabi and Salem, 2025; Laine et al., 2024; Greenblatt et al., 2024). However, three questions remain open. First, how strong and widespread is this phenomenon in open models rather than only frontier systems? Second, when evaluation awareness is present, does it actually change safety-relevant behavior? Third, does “evaluation awareness” refer to a single underlying capability, or do different benchmarks measure different and only partially related aspects of the phenomenon? These questions matter because deployment decisions are often made from benchmark evidence, yet benchmark validity depends on whether models treat evaluation and deployment equivalently.

We present a comprehensive empirical study of behavioral manifestations of evaluation awareness, spanning eight experiments across 37 open models from seven families. By analyzing base, instruction-tuned, and reasoning variants using linear probes, sparse autoencoders, activation steering, and tiered real-vs-hypothetical framing, we organize the paper around four research questions. We ask how open models detect evaluation context (**RQ₁**), whether this detection shifts safety behavior (**RQ₂**), whether it reflects genuine awareness or format artifacts (**RQ₃**), and whether it can be localized and causally intervened upon (**RQ₄**).

Four findings drive the paper:

1. An **independent replication** of evaluation awareness on 37 open models and independent infrastructure, including a benign control that validates the pipeline.
2. The first **family/scale/training-type detection law**, showing that evaluation awareness

is driven primarily by instruction tuning rather than raw scale, with no reliable gain from reasoning modes.

3. A **causal mechanistic account** showing that evaluation-awareness signals survive rewrite controls and can be causally manipulated through activation steering.
4. A **cross-experiment synthesis** showing that evaluation awareness is not one trait but at least **three weakly-coupled axes** (only 1/15 paradigm correlations significant), with a single robust, *size-independent* coupling between behavioral detection and resistance to safety-framing ($\rho = -0.79$).

Our central claim is not that models are merely evaluation-aware, but that benchmark-based safety claims can be misleading unless they specify which axis of awareness they measure. We show that detection, behavior, and controllability can diverge substantially, creating a benchmark illusion: a gap between benchmark performance and deployment behavior.

2 Related Work

Benchmarks and datasets for evaluation awareness. Prior work studies evaluation awareness through transcript-classification and situational-awareness benchmarks (Needham et al. (2025); Laine et al. (2024)). Recent work further shows that benchmark conclusions can depend strongly on prompt wording and surface form, motivating rewrite controls and cross-format evaluation (Xiong et al., 2025; Mizrahi et al., 2024).

Artifacts vs. Genuine Awareness. A central concern is that apparent evaluation awareness may reflect sensitivity to benchmark artifacts rather than genuine context recognition. Prior work demonstrates substantial dependence on prompt structure and surface cues (Zheng et al., 2024; Balepur et al., 2024), motivating the artifact controls used in our study. Devbunova (2026) shows that standard single-contrast probes trained on benchmark-vs-chat data primarily track prompt format rather than evaluation context, and that a paired training design decorrelating format from context recovers a genuine context signal.

Mechanistic Analysis. Recent interpretability work shows that behaviorally relevant signals can often be localized and manipulated through probing and activation interventions (Li et al., 2023;

Zou et al., 2023; Ardit et al., 2024). We adopt these methods, together with SAE-based analysis (Lieberum et al., 2024), to study the internal representation of evaluation awareness, building on recent work applying probing and steering techniques to evaluation-awareness representations (Nguyen et al., 2025; Hua et al., 2025).

3 Research Methodology

3.1 Conceptual Framework

We study evaluation awareness across three levels. *Behavioral evaluation awareness* refers to whether evaluation-like or deployment-like framing changes model outputs, for example, increasing refusal during a safety evaluation or increasing compliance when a harmful request is framed as hypothetical. *Representational evaluation awareness* refers to whether hidden states encode the eval/deploy distinction independent of behavioral output. *Mechanistic evaluation awareness* refers to whether specific internal features (SAE directions, activation steering vectors) causally mediate the signal and can be intervened upon. These three levels can dissociate: a model may encode the distinction without changing behavior, behavior may shift due to shallow formatting cues without a reusable internal representation, and a representation may be detectable without being causally actionable.

To operationalize **RQ₂**, our central measurement device is the Safety Validity Gap (SVG):

$$\text{SVG}(S, c_1, c_2) = |S_{c_1} - S_{c_2}| \quad (1)$$

where S is a safety-relevant metric and c_1, c_2 are semantically matched contexts differing in evaluation-likeness. We instantiate SVG in three concrete ways: (1) real-vs-hypothetical refusal SVG in the Hawthorne experiment, (2) compliance SVG across HarmBench framing tiers, and (3) a control SVG on harmless prompts where the expected gap is near zero, validating pipeline integrity.

3.2 Research Questions

We organize the study around four questions: whether models detect evaluation context, whether such detection alters safety-relevant behavior, whether benchmark signals reflect genuine awareness or format artifacts, and whether the underlying representation can be localized and causally manipulated. Tab. 1 summarizes, for each research

Table 1: Per-research-question model scope, rationale, and methods. The model count is deliberately question-dependent: **RQ₁** spans the full generator diversity, whereas **RQ₄** narrows to models with accessible internals and SAE tooling.

Research Question	Methods	Models & Rationale
RQ₁ : To what extent do open models distinguish evaluation-like from deployment-like contexts, and how does this vary across family, scale, and training type?	Needham-style transcript classification, bootstrap significance tests, FDR correction, pairwise AUC comparison	37 models (0.5B-32B) across 7 families, base/instruct/reasoning variants - full diversity needed to characterize the detection landscape
RQ₂ : Holding task semantics constant, to what extent does framing a prompt as evaluation versus deployment cause a measurable shift in a model’s safety-relevant behavior?	Real-vs-hypothetical Hawthorne pairs, HarmBench 5-tier framing, SVG computation, benign null control	35 models for HarmBench tiered framing; single model (DeepSeek-R1-Distill-Llama-8B) for Hawthorne - framing effects established on a focused case before scaling
RQ₃ : To what extent do observed detection effects reflect genuine context recognition versus sensitivity to superficial formatting artifacts such as MCQ structure, benchmark phrasing, explicit test headers, or sterile prompt register?	SAD stages-oversight rewrite tiers, system-prompt rewrite, 2×2 benchmark/casual × eval/deploy factorial design, surface-feature classifier baselines	32-36 models for SAD and 2×2 controlled design - broad sweep confirms artifact sensitivity is systematic rather than model-specific
RQ₄ : Which hidden states, steering directions, or SAE features encode evaluation-awareness signals, and to what extent do interventions on those representations causally affect downstream safety behavior?	Layer-wise linear probing, SAE cosine alignment, mean activation differential, activation steering, benign capability retention checks	Gemma-3-1B-IT and Gemma-3-4B-IT prioritized - mechanistic analysis requires models with accessible internals and compatible SAE tooling (Gemma Scope)

question, the model scope, the rationale for that scope, and the methods employed.

RQ₁: [Detection] We treat detection as a per-model classification problem and run Needham-style transcript classification on all 37 models (0.5B-32B, seven families, base, instruction-tuned, and reasoning variants). The full sweep is deliberate: characterizing *which* models detect evaluation and *why* requires spanning the diversity of generators, so this is the one question that uses the entire model pool. We score each model with *AU-ROC*, establish significance with transcript-level bootstrap resampling, and compare models with FDR-corrected pairwise *AUROC* tests. Base-vs-instruct and thinking-vs-non-thinking contrasts isolate the training-type effect.

RQ₂: [Behavioral Consequence] We hold task semantics fixed and vary only evaluation-likeness, then measure the resulting shift in safety behavior with the Safety Validity Gap (Eq. 1). Two complementary instruments are used: real-vs-hypothetical Hawthorne pairs run as a focused single-model case (DeepSeek-R1-Distill-Llama-8B, 2,587 pairs) to establish the effect cleanly before scaling, and HarmBench five-tier framing across 35 models to test generality. A benign, byte-identical null control bounds harness noise, so any nonzero SVG is

attributable to framing rather than the pipeline.

RQ₃: [Artifact vs. Awareness] To decide whether detection reflects context recognition or prompt-format sensitivity, we apply controlled perturbations that preserve meaning while varying surface form: SAD stages-oversight rewrite tiers, an explicit “being tested” system-prompt rewrite, and a 2×2 benchmark/casual × eval/deploy factorial that lets us transplant format across contexts, complemented by MCQ/surface-feature baselines. The broad 32-36 model sweep here is intended to show that artifact sensitivity is systematic rather than model-specific.

RQ₄: [Mechanistic Representation & Causal Role] We move from behavior to mechanism in three steps. Layer-wise linear probes test where the eval/deploy signal is decodable in the residual stream; SAE analysis (Gemma Scope) decomposes the probe direction via cosine alignment and mean-activation differential to ask whether it corresponds to interpretable features or to formatting artifacts; and activation steering tests whether intervening on those representations causally changes downstream safety behavior, paired with benign capability-retention checks. This arm prioritizes Gemma-3-1B-IT and Gemma-3-4B-IT because mechanistic analysis requires accessible internals and compati-